**Database Assignment 1**

**Schema Design & Table Creation (DDL & DML Commands)**

1. Define a schema for a Library Management System with the following entities:
   - Books
   - Authors
   - Members
   - Borrow_Records

Code:

```
CREATE DATABASE LibraryDB;

USE LibraryDB;

CREATE TABLE Authors (
    author_id INT PRIMARY KEY,
    name VARCHAR(100),
    country VARCHAR(50)
);

CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(150),
    author_id INT,
    published_year YEAR,
    available_copies INT,
    FOREIGN KEY (author_id) REFERENCES Authors(author_id)
);

CREATE TABLE Members (
    member_id INT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100) UNIQUE,
    phone VARCHAR(15)
);

CREATE TABLE Borrow_Records (
    record_id INT PRIMARY KEY,
    member_id INT,
```

```
    book_id INT,
    borrow_date DATE,
    return_date DATE,
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
```

Run these commands:

```
SHOW DATABASES;
USE LibraryDB;
SHOW TABLES;
```

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| aug2026            |
| information_schema |
| librarydb          |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)

mysql>
mysql> USE LibraryDB;
Database changed
mysql>
mysql> SHOW TABLES;
+---------------------+
| Tables_in_librarydb |
+---------------------+
| authors             |
| books               |
| borrow_records      |
| members             |
+---------------------+
4 rows in set (0.01 sec)
```

2. Write the SQL command to create a table Authors with the following fields:
   - author_id (Primary Key, INT)
   - name (VARCHAR(100))
   - country (VARCHAR(50))

Code:

```
CREATE TABLE Authors (
```

```
    author_id INT PRIMARY KEY,
    name VARCHAR(100),
    country VARCHAR(50)
);
DESC Authors;
```

```
mysql> DESC Authors;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| author_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| country   | varchar(50)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
3 rows in set (0.02 sec)
```

3. Write the SQL command to create a table Books with the following fields:
   - book_id (Primary Key, INT)
   - title (VARCHAR(150))
   - author_id (Foreign Key referencing Authors)
   - published_year (YEAR)
   - available_copies (INT)

Code:

```
USE LibraryDB;
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(150),
    author_id INT,
    published_year YEAR,
    available_copies INT,
    FOREIGN KEY (author_id) REFERENCES Authors(author_id)
);
DESC Books;
```

```
mysql> DESC Books;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| book_id         | int         | NO   | PRI | NULL    |       |
| title           | varchar(150)| YES  |     | NULL    |       |
| author_id       | int         | YES  | MUL | NULL    |       |
| published_year  | year        | YES  |     | NULL    |       |
| available_copies| int         | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

4. Write the SQL command to create a table Members with:
   - member_id (Primary Key, INT)
   - name (VARCHAR(100))
   - email (VARCHAR(100), unique)
   - phone (VARCHAR(15))

Code:

```
USE LibraryDB;

CREATE TABLE Members (

   member_id INT PRIMARY KEY,

   name VARCHAR(100),

   email VARCHAR(100) UNIQUE,

   phone VARCHAR(15)

);

USE LibraryDB;

DESC Members;
```

```
mysql> USE LibraryDB;
Database changed
mysql> DESC Members;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| member_id | int         | NO   | PRI | NULL    |       |
| name      | varchar(100)| YES  |     | NULL    |       |
| email     | varchar(100)| YES  | UNI | NULL    |       |
| phone     | varchar(15) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
4 rows in set (0.04 sec)
```

5. Write the SQL command to create a table Borrow_Records with:
- record_id (Primary Key, INT)
- member_id (Foreign Key referencing Members)
- book_id (Foreign Key referencing Books)
- borrow_date (DATE)
- return_date (DATE)

Code:

```
USE LibraryDB;

CREATE TABLE Borrow_Records (

   record_id INT PRIMARY KEY,

   member_id INT,

   book_id INT,

   borrow_date DATE,

   return_date DATE,

   FOREIGN KEY (member_id) REFERENCES Members(member_id),

   FOREIGN KEY (book_id) REFERENCES Books(book_id)

);

USE LibraryDB;

DESC Borrow_Records;
```

```
mysql> USE LibraryDB;
Database changed
mysql> DESC Borrow_Records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  | MUL | NULL    |       |
| book_id     | int  | YES  | MUL | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

6. Modify the Books table to add a column genre of type VARCHAR(50).

```
USE LibraryDB;

ALTER TABLE Books
```

ADD genre VARCHAR(50);

USE LibraryDB;

DESC Books;

```
mysql> USE LibraryDB;
Database changed
mysql> DESC Books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
| genre            | varchar(50)  | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

7. Write the SQL command to drop the Borrow_Records table.

DROP TABLE Borrow_Records;

SHOW TABLES;

```
mysql> DROP TABLE Borrow_Records;
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+---------------------+
| Tables_in_librarydb |
+---------------------+
| authors             |
| books               |
| members             |
+---------------------+
3 rows in set (0.01 sec)
```

8. Insert 3 records into the Authors table.

INSERT INTO Authors (author_id, name, country)

VALUES

(1, 'Ramesh Sharma', 'India'),

(2, 'John Smith', 'USA'),

(3, 'Priya Patel', 'India');

USE LibraryDB;

SELECT * FROM Authors;

```
mysql> USE LibraryDB;
Database changed
mysql>
mysql> SELECT * FROM Authors;
+-----------+---------------+---------+
| author_id | name          | country |
+-----------+---------------+---------+
|         1 | Ramesh Sharma | India   |
|         2 | John Smith    | USA     |
|         3 | Priya Patel   | India   |
+-----------+---------------+---------+
3 rows in set (0.00 sec)
```

9. Insert 5 books into the Books table.

INSERT INTO Books

(book_id, title, author_id, published_year, available_copies, genre)

VALUES

(1, 'Database Systems', 1, 2018, 5, 'Education'),

(2, 'Java Programming', 2, 2020, 3, 'Programming'),

(3, 'Python Basics', 3, 2019, 4, 'Programming'),

(4, 'Computer Networks', 1, 2017, 6, 'Technology'),

(5, 'Data Structures', 2, 2022, 2, 'Education');

USE LibraryDB;

SELECT * FROM Boks;

```
mysql> USE LibraryDB;
Database changed
mysql>
mysql> SELECT * FROM Books;
+---------+------------------+-----------+----------------+----------------
--+-------------+
| book_id | title            | author_id | published_year | available_copie
s | genre       |
+---------+------------------+-----------+----------------+----------------
--+-------------+
|       1 | Database Systems |         1 |           2018 |
5 | Education   |
|       2 | Java Programming |         2 |           2020 |
3 | Programming |
|       3 | Python Basics    |         3 |           2019 |
4 | Programming |
|       4 | Computer Networks |         1 |           2017 |
6 | Technology  |
|       5 | Data Structures  |         2 |           2022 |
2 | Education   |
+---------+------------------+-----------+----------------+----------------
--+-------------+
5 rows in set (0.01 sec)
```

10. Insert 3 members into the Members table.

```
INSERT INTO Members (member_id, name, email, phone)
VALUES
(1, 'Amit Patil', 'amit@gmail.com', '9876543210'),
(2, 'Sneha Sharma', 'sneha@gmail.com', '9876543211'),
(3, 'Rahul Verma', 'rahul@gmail.com', '9876543212');
USE LibraryDB;
SELECT * FROM Members;
```

```
mysql> USE LibraryDB;
Database changed
mysql>
mysql> SELECT * FROM Members;
+-----------+--------------+-----------------+------------+
| member_id | name         | email           | phone      |
+-----------+--------------+-----------------+------------+
|         1 | Amit Patil   | amit@gmail.com  | 9876543210 |
|         2 | Sneha Sharma | sneha@gmail.com | 9876543211 |
|         3 | Rahul Verma  | rahul@gmail.com | 9876543212 |
+-----------+--------------+-----------------+------------+
3 rows in set (0.00 sec)
```

11. Insert 4 borrow records into the Borrow_Records table.

```
INSERT INTO Borrow_Records
(record_id, member_id, book_id, borrow_date, return_date)
VALUES
(1, 1, 1, '2026-10-01', '2026-10-05'),
(2, 2, 2, '2026-10-02', '2026-10-06'),
(3, 3, 3, '2026-10-03', NULL),
(4, 1, 4, '2026-10-04', '2026-10-08');
SELECT * FROM Borrow_Records;
```

```
mysql> INSERT INTO Borrow_Records
    -> (record_id, member_id, book_id, borrow_date, return_date)
    -> VALUES
    -> (1, 1, 1, '2026-10-01', '2026-10-05'),
    -> (2, 2, 2, '2026-10-02', '2026-10-06'),
    -> (3, 3, 3, '2026-10-03', NULL),
    -> (4, 1, 4, '2026-10-04', '2026-10-08');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Borrow_Records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|         1 |         1 |       1 | 2026-10-01  | 2026-10-05  |
|         2 |         2 |       2 | 2026-10-02  | 2026-10-06  |
|         3 |         3 |       3 | 2026-10-03  | NULL        |
|         4 |         1 |       4 | 2026-10-04  | 2026-10-08  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.01 sec)
```

12. Write an SQL query to select all books where published_year is after 2015.

SELECT *

FROM Books

WHERE published_year > 2015;

```
mysql>
mysql> SELECT *
    -> FROM Books
    -> WHERE published_year > 2015;
+---------+------------------+-----------+----------------+------------------+-------------+
| book_id | title            | author_id | published_year | available_copies | genre       |
+---------+------------------+-----------+----------------+------------------+-------------+
|       1 | Database Systems |         1 |           2018 |                5 | Education   |
|       2 | Java Programming |         2 |           2020 |                3 | Programming |
|       3 | Python Basics    |         3 |           2019 |                4 | Programming |
|       4 | Computer Networks|         1 |           2017 |                6 | Technology  |
|       5 | Data Structures  |         2 |           2022 |                2 | Education   |
+---------+------------------+-----------+----------------+------------------+-------------+
5 rows in set (0.01 sec)
```

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

CREATE TABLE Authors (

  author_id INT PRIMARY KEY,

  name VARCHAR(100),

  country VARCHAR(50)

);

```
CREATE TABLE Books (
   book_id INT PRIMARY KEY,
   title VARCHAR(150),
   author_id INT,
   FOREIGN KEY (author_id)
      REFERENCES Authors(author_id)
);
```

SHOW CREATE TABLE Books;

```
mysql> SHOW CREATE TABLE Books;
+-------+-----------------------
```

```
mysql>
mysql> SELECT Members.member_id, Members.name, Members.email
    -> FROM Members
    -> JOIN Borrow_Records
    -> ON Members.member_id = Borrow_Records.member_id
    -> JOIN Books
    -> ON Borrow_Records.book_id = Books.book_id
    -> WHERE Books.title = 'Database Systems';
+-----------+------------+----------------+
| member_id | name       | email          |
+-----------+------------+----------------+
|         1 | Amit Patil | amit@gmail.com |
+-----------+------------+----------------+
1 row in set (0.01 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

UPDATE Books

SET available_copies = available_copies - 1

WHERE book_id = 1;

SELECT book_id, title, available_copies

FROM Books

WHERE book_id = 1;

```
mysql> UPDATE Books
    -> SET available_copies = available_copies - 1
    -> WHERE book_id = 1;
Query OK, 1 row affected (0.10 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT book_id, title, available_copies
    -> FROM Books
    -> WHERE book_id = 1;
+---------+-----------------+------------------+
| book_id | title           | available_copies |
+---------+-----------------+------------------+
|       1 | Database Systems |                4 |
+---------+-----------------+------------------+
1 row in set (0.01 sec)
```

16. Delete a record from Members where member_id = 3.

SELECT * FROM Members

WHERE member_id = 3;

```
mysql> SELECT * FROM Members
    -> WHERE member_id = 3;
+-----------+-------------+-----------------+------------+
| member_id | name        | email           | phone      |
+-----------+-------------+-----------------+------------+
|         3 | Rahul Verma | rahul@gmail.com | 9876543212 |
+-----------+-------------+-----------------+------------+
1 row in set (0.00 sec)
```

17. Update a Book name record from Book table  with id = 1.

UPDATE Books

SET title = 'SQL Database Systems'

WHERE book_id = 1;

SELECT * FROM Books

WHERE book_id = 1;

```
mysql> UPDATE Books
    -> SET title = 'SQL Database Systems'
    -> WHERE book_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Books
    -> WHERE book_id = 1;
+---------+---------------------+-----------+----------------+------------------+-----------+
| book_id | title               | author_id | published_year | available_copies | genre     |
+---------+---------------------+-----------+----------------+------------------+-----------+
|       1 | SQL Database Systems |         1 |           2018 |                4 | Education |
+---------+---------------------+-----------+----------------+------------------+-----------+
1 row in set (0.00 sec)
```

18. Write an SQL query to list all books along with their authors' names.

SELECT Books.book_id, Books.title, Authors.name

FROM Books

JOIN Authors

ON Books.author_id = Authors.author_id;

```
mysql> SELECT Books.book_id, Books.title, Authors.name
    -> FROM Books
    -> JOIN Authors
    -> ON Books.author_id = Authors.author_id;
+---------+----------------------+---------------+
| book_id | title                | name          |
+---------+----------------------+---------------+
|       1 | SQL Database Systems | Ramesh Sharma |
|       4 | Computer Networks    | Ramesh Sharma |
|       2 | Java Programming     | John Smith    |
|       5 | Data Structures      | John Smith    |
|       3 | Python Basics        | Priya Patel   |
+---------+----------------------+---------------+
5 rows in set (0.05 sec)
```

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

DELETE FROM Books

WHERE published_year < 2000;

SELECT * FROM Books;

```
mysql> DELETE FROM Books
    -> WHERE published_year < 2000;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Books;
+---------+---------------------+-----------+----------------+------------------+-------------+
| book_id | title               | author_id | published_year | available_copies | genre       |
+---------+---------------------+-----------+----------------+------------------+-------------+
|       1 | SQL Database Systems |         1 |           2018 |                4 | Education   |
|       2 | Java Programming     |         2 |           2020 |                3 | Programming |
|       3 | Python Basics        |         3 |           2019 |                4 | Programming |
|       4 | Computer Networks    |         1 |           2017 |                6 | Technology  |
|       5 | Data Structures      |         2 |           2022 |                2 | Education   |
+---------+---------------------+-----------+----------------+------------------+-------------+
5 rows in set (0.00 sec)
```

20. Write an SQL query to **find all books that are never borrowed (i.e., no records in Borrow_Records).**

SELECT Books.*

FROM Books

LEFT JOIN Borrow_Records

ON Books.book_id = Borrow_Records.book_id

WHERE Borrow_Records.book_id IS NULL;

```
mysql>
mysql> SELECT Books.*
    -> FROM Books
    -> LEFT JOIN Borrow_Records
    -> ON Books.book_id = Borrow_Records.book_id
    -> WHERE Borrow_Records.book_id IS NULL;
+---------+----------------+-----------+----------------+------------------+-----------+
| book_id | title          | author_id | published_year | available_copies | genre     |
+---------+----------------+-----------+----------------+------------------+-----------+
|       5 | Data Structures |         2 |           2022 |                2 | Education |
+---------+----------------+-----------+----------------+------------------+-----------+
1 row in set (0.01 sec)
```